HDOC-14

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Program to Print the Sum of the First n Odd Numbers

This document provides a clean, well-commented implementation for finding and printing the sum of the first **n** odd numbers in C. The program takes the value of **n** as input and uses a for loop to calculate the cumulative sum efficiently.

1. C Implementation

Step 1: Understand the Logic

The first **n** odd numbers are 1, 3, 5, ..., up to the n-th odd number (2n − 1). In a loop running **n** times (i from 1 to **n**), the i-th odd number is given by 2i − 1, which is added to a running total (`sum`).

- **Initialization:** `sum = 0` and i = 1, so counting starts from the first odd number.
- **Condition:** i ≤ n, so the loop runs exactly **n** times.
- **Update:** `i++`, so the counter increases by 1 after every pass.
- If **n** is less than 1, the input is rejected with an error message.

Step 2: Define Variables and Data Types

Use `int` variables for the limit (`n`), the loop counter (`i`), and the accumulated sum (`sum`).

Step 3: Take User Input

Display a prompt using `printf("Enter the value of n: ");`, then read the value using `scanf("%d", &n);`.

Step 4: Validate the Input

If n < 1, print an error message and do not run the loop.

Step 5: Calculate and Print the Sum Using a for Loop

1. Start with i = 1 and `sum = 0`.
2. While i ≤ n, add (2i − 1) to `sum`.
3. Increase i by 1 and check the condition again.
4. When i becomes greater than **n**, the loop ends. Print the final calculated sum.

Complete C Code

```
/* * Q22 (User Inputs, Operations & Output) * Write a program to print the sum of the first n odd numbers. */ #include <stdio.h> int main() { int n, i, sum = 0; // Step 1: Read the value of n printf("Enter the value of n: "); scanf("%d", &n); // Step 2: Validate n if (n < 1) { printf("Invalid input! n must be at least 1.\n"); } else { // Step 3: Calculate sum using a for loop for (i = 1; i <= n; i++) { sum += (2 * i - 1); } printf("The sum of the first %d odd numbers is: %d\n", n, sum); } return 0; }
```

Program Output (Sample Runs)

```
Run 1: $ gcc hdoc14.c -o hdoc14 $ ./hdoc14 Enter the value of n: 3 The sum of the first 3 odd numbers is: 9 Run 2: $ ./hdoc14 Enter the value of n: 5 The sum of the first 5 odd numbers is: 25 Run 3: $ ./hdoc14 Enter the value of n: 1 The sum of the first 1 odd numbers is: 1 Run 4: $ ./hdoc14 Enter the value of n: 0 Invalid input! n must be at least 1.
```

Dry Run with Test Cases

Sample Test Case 1:
Input: n = 3
Calculation:

- 3 < 1 is FALSE, so the input is valid and the loop runs.
- Start: `i = 1`, `sum = 0`.
- Pass 1: 1 ≤ 3 is TRUE, add 2(1) − 1 = 1 to sum (`sum = 1`), i becomes 2.
- Pass 2: 2 ≤ 3 is TRUE, add 2(2) − 1 = 3 to sum (`sum = 4`), i becomes 3.
- Pass 3: 3 ≤ 3 is TRUE, add 2(3) − 1 = 5 to sum (`sum = 9`), i becomes 4.
- Check: 4 ≤ 3 is FALSE, so the loop ends.

Output: The sum of the first 3 odd numbers is: 9

Sample Test Case 2:
Input: n = 1
Calculation:

- 1 < 1 is FALSE, valid input. Start: `i = 1`, `sum = 0`.
- Pass 1: 1 ≤ 1 is TRUE, add 2(1) − 1 = 1 to sum (`sum = 1`), i becomes 2.
- Check: 2 ≤ 1 is FALSE, loop ends after one pass.

Output: The sum of the first 1 odd numbers is: 1

Sample Test Case 3:
Input: n = 0
Calculation: 0 < 1 is TRUE, input is invalid. Error message is printed and loop does not run.
Output: Invalid input! n must be at least 1.